

A Scalar-Shift Counterexample to the Local Bishop- β Sum Question

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Source and Question

The source paper is Salah Mecheri, *On the stability of the Bishop's property(β) under compact perturbations*, arXiv:2510.09289. On page 5, immediately after Theorem 2.1, it asks the following localized question:

If $S, T \in B(X)$ have Bishop's property (β) at λ_0 , does $S + T$ have Bishop's property (β) at λ_0 ?

that $f_n(\lambda) \rightarrow 0$ as $n \rightarrow \infty$ uniformly on all compact subsets of U .
Hence, S has the Bishop's property β . $\square$

If $S, T \in B(X)$ have Bishop's property (β) at λ_0 , does $S + T$ have Bishop's property (β) at λ_0 ? So far we do not know the answer about this question. In the following theorem we study a special case of this question.

Theorem 2.2. *If R has Bishop's property (β) at λ_0 , $S \in \text{Helton}_k(R)$ and $RS = SR$, then $T = R + S$ has Bishop's property (β) at λ_0 .*

Proof. It is easily seen that $C(2R; S)^k(I) = C(R; S)^k(I) = 0$. Hence $T = R + S \in \text{Helton}_k(2R)$. Since $2R$ has Bishop's property (β) it

Figure 1: The localized sum question in arXiv:2510.09289, p. 5.

The answer to this question, as stated with the same fixed base point λ_0 , is negative.

Definitions

Let $A \in \mathcal{B}(X)$, where X is a complex Banach space. We say that A has Bishop's property (β) at λ_0 if there is $r > 0$ such that, for every open set $U \subset D(\lambda_0, r)$ and every sequence $(f_n) \subset \mathcal{O}(U; X)$,

$$(A - \lambda I)f_n(\lambda) \longrightarrow 0 \quad \text{in } \mathcal{O}(U; X)$$

implies

$$f_n(\lambda) \longrightarrow 0 \quad \text{in } \mathcal{O}(U; X).$$

The single-valued extension property, abbreviated SVEP, at λ_0 says that the only analytic f near λ_0 with $(A - \lambda I)f(\lambda) = 0$ is $f \equiv 0$.

Proof Intuition

The localized formulation is not stable under scalar translation at a fixed base point. Start from an operator B that fails Bishop's property (β) at 0. Shift it far away to $B - 2I$; now 0 lies in the resolvent set, so the shifted operator has (β) at 0. The scalar operator $2I$ also has (β) at 0. But adding the two commuting operators gives back B , which was chosen to fail (β) at 0.

The backward unilateral shift on ℓ^2 supplies the required B : it has a nonzero analytic eigenvector field $z \mapsto (1, z, z^2, \dots)$ for $|z| < 1$, so it fails SVEP, and hence fails Bishop's property (β) , at the origin.

The Counterexample

Lemma 1. *If $\lambda_0 \in \rho(A)$, then A has Bishop's property (β) at λ_0 .*

Proof. Choose $r > 0$ such that $\overline{D(\lambda_0, r)} \subset \rho(A)$. Let $U \subset D(\lambda_0, r)$ be open and suppose $(A - \lambda I)f_n(\lambda) \rightarrow 0$ in $\mathcal{O}(U; X)$. For each compact $K \subset U$, the resolvent is uniformly bounded on K , so

$$\sup_{\lambda \in K} \|f_n(\lambda)\| \leq \sup_{\lambda \in K} \|(A - \lambda I)^{-1}\| \sup_{\lambda \in K} \|(A - \lambda I)f_n(\lambda)\| \rightarrow 0.$$

Thus $f_n \rightarrow 0$ in $\mathcal{O}(U; X)$. □

Lemma 2. *If A has Bishop's property (β) at λ_0 , then A has SVEP at λ_0 .*

Proof. If f is a nonzero analytic solution of $(A - \lambda I)f(\lambda) = 0$ on a neighborhood of λ_0 , then the constant sequence $f_n = f$ satisfies $(A - \lambda I)f_n(\lambda) \equiv 0$, but f_n does not converge to 0. This contradicts Bishop's property (β) . □

Theorem 1. *There are commuting operators $S, T \in \mathcal{B}(\ell^2(\mathbb{N}_0))$ such that both S and T have Bishop's property (β) at 0, but $S + T$ does not have Bishop's property (β) at 0.*

Proof. Let B be the backward unilateral shift on $\mathcal{H} = \ell^2(\mathbb{N}_0)$:

$$B(x_0, x_1, x_2, \dots) = (x_1, x_2, x_3, \dots).$$

Define

$$S = 2I, \quad T = B - 2I.$$

Plainly S and T commute.

The operator S has Bishop's property (β) at 0, since $0 \in \rho(S)$. The operator T also has Bishop's property (β) at 0. Indeed, for $|\lambda| < 1/2$,

$$T - \lambda I = B - (2 + \lambda)I$$

is invertible by the Neumann series, because

$$\left\| \frac{B}{2 + \lambda} \right\| \leq \frac{1}{|2 + \lambda|} < 1.$$

Hence $0 \in \rho(T)$, and Lemma 1 applies.

It remains to show that $S + T = B$ does not have Bishop's property (β) at 0. For $|z| < 1$, put

$$g(z) = (1, z, z^2, \dots) \in \ell^2(\mathbb{N}_0).$$

The map $g : D(0, 1) \rightarrow \ell^2(\mathbb{N}_0)$ is analytic and nonzero, and

$$Bg(z) = (z, z^2, z^3, \dots) = z g(z).$$

Thus

$$(B - zI)g(z) = 0 \quad (|z| < 1),$$

so B fails SVEP at 0. By Lemma 2, B cannot have Bishop's property (β) at 0. Since $S + T = B$, this proves the claim. $\square$

Scope and Limitations

This is a counterexample to the localized question exactly as printed on page 5 of the source paper: both summands have Bishop's property (β) at the same fixed point 0, while their sum fails it at that same point.

It does not settle the broader global problem discussed in the source introduction, namely whether sums or products of commuting operators that have Bishop's property (β) everywhere must again have Bishop's property (β) everywhere. In the example above, $T = B - 2I$ is used only through its local behavior at 0; no global Bishop- β assertion is made for T .

Novelty Check

The run's cheap indexes were searched for arXiv:2510.09289, the source title, "Bishop's property", "Bishop beta", "localized Bishop", "SVEP", "commuting perturbation", and "backward shift". No prior packet, proof gap, or attempt for this exact localized question was found.

A bounded web search for the exact source-question phrase and nearby phrases found the source preprint, the older/global formulation in Mecheri's 2012 paper, and standard background statements such as Bishop's property (β) implying SVEP. It did not reveal an existing scalar-shift counterexample to the fixed-base-point localized sentence. This is not an exhaustive literature search; because the example is elementary, it may be folklore. The mathematical claim above is nevertheless self-contained.

Verification Notes and Human Review

Recommended review verdict: likely valid counterexample to the localized question. The key checks are:

- verify that the source question asks for the sum at the same fixed λ_0 ;
- verify Lemma 1, especially the use of the resolvent neighborhood for a local property;
- verify that $z \mapsto (1, z, z^2, \dots)$ is a nonzero analytic ℓ^2 -valued solution of $(B - zI)g(z) = 0$ near 0.

No computational verification is needed.

References

- [1] Salah Mecheri, *On the stability of the Bishop's property(β) under compact perturbations*, arXiv:2510.09289, 2025.
- [2] K. B. Laursen and M. M. Neumann, *An Introduction to Local Spectral Theory*, London Mathematical Society Monographs, Clarendon Press, Oxford, 2000.